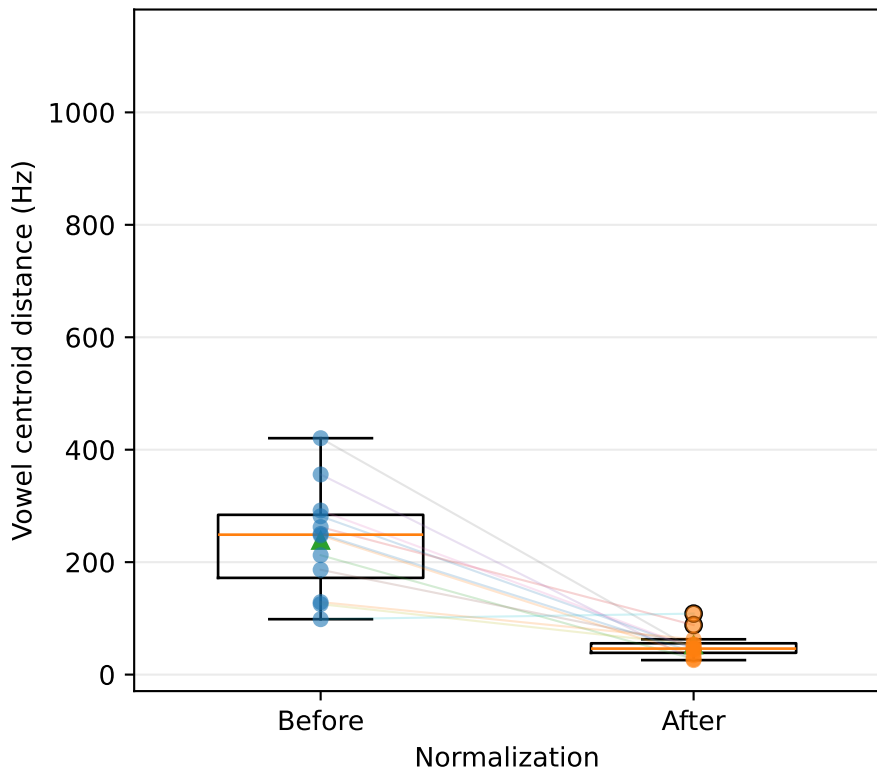
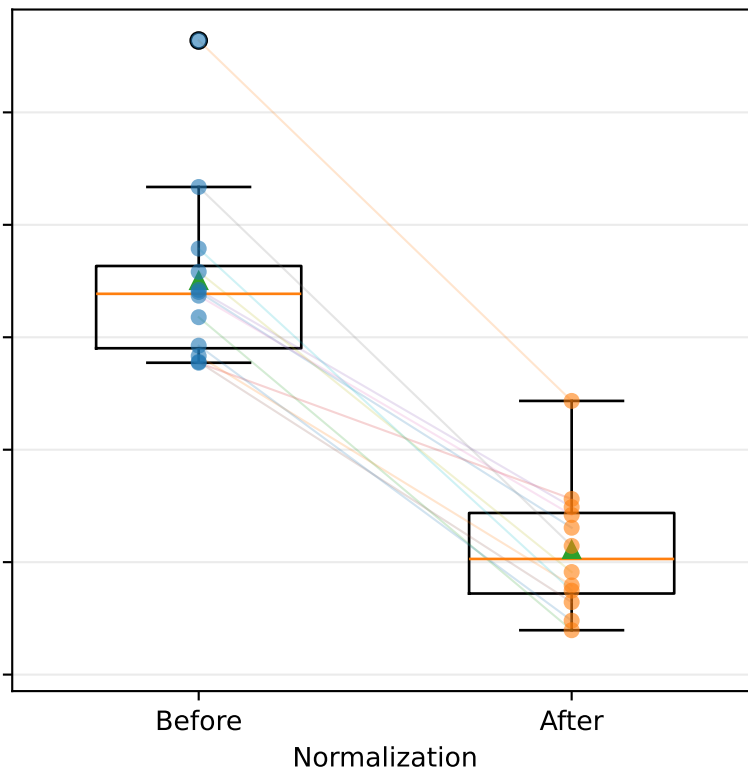


Cross-Group Vowel Centroid Distances Before and After VTL Normalization

Men vs. Women
238.5 → 52.6 Hz (77.9% reduction)



Men vs. Children
700.5 → 222.1 Hz (68.3% reduction)



Women vs. Children
468.8 → 231.6 Hz (50.6% reduction)

